TED UNIVERSITY

Department of Computer Engineering | CMPE 491

SENIOR DESIGN PROJECT I
PRODUCT BACKLOG

ReviewShield: A Multi-Signal Framework for Detecting Coordinated Fake Reviews in Turkish E-Commerce

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Backlog version	Version 1 (D3) – 23 October 2026
Project webpage	https://aliyagizt.github.io/ReviewShield/

1. Backlog Overview

This Product Backlog outlines the tasks required to develop ReviewShield based on the Project Specifications Report. It organizes the planned work into manageable items covering data preparation, text similarity analysis, temporal analysis, graph analysis, suspicion scoring, the API and dashboard, system evaluation and the course deliverables. Each item includes a priority, its source in the Project Specifications Report, acceptance criteria, a planned sprint, its current status and the responsible team member.

1.1 Legend

Table 1. Fields used in the backlog

Field	**Meaning**
Priority	**Must** (part of the minimum viable prototype), **Should** (planned, but the prototype remains usable without it), **Could** (stretch goal) and **Conditional** (required only if its stated condition occurs).
Source	Section, functional requirement (FR) or non-functional requirement (NFR) of the Project Specifications Report, or course deliverable (D1–D8).
Sprint	Planned sprint (see Table 2). “S1–S3” marks continuous work across all sprints; “If time” marks optional stretch goals.
Status	**Not started**, **In progress** or **Done**.
Assignee	Team member responsible for the item (Sinem Şahin, Gizem Demirer, Ali Yağız Temizhan or Bora Özkan).

1.2 Sprint Plan

Development starts with dataset selection, validation and preparation, continues with the analysis modules and group-level scoring, and ends with integration into the web dashboard, testing and evaluation. Tasks can be developed in parallel once the data format is agreed upon.

Table 2. Sprint plan

Sprint	Period	Sprint goal	Course deliverables
S1	23 Oct – 20 Nov 2026	Dataset audit and selection; input schema, import and validation; Turkish preprocessing; text embeddings and similarity; temporal aggregation and burst detection.	D4 Analysis Report; D5 Backlog v2 (20 Nov)
S2	21 Nov – 31 Dec 2026	Near-simultaneous activity signal; reviewer–product graph and candidate groups; group scoring and evidence; API and dashboard views.	D6 High-Level Design Report; D7 Backlog v3 (31 Dec)
S3	1 Jan 2027 – demonstration	Integration, graph view, evaluation configurations and metrics, error analysis, end-to-end analyst workflow and demonstration.	D8 Presentation, poster and demonstration (4–17 Jan 2027)

2. Product Backlog

A. Data access, import and preprocessing

ID	Backlog item	Priority	Source	Acceptance criteria	Sprint	Status	Assignee
PB-01	Audit candidate datasets for provenance, permissions, language and field completeness.	Must	§1.2; NFR-04	Dataset source and access date recorded; terms/license and permitted use reviewed; available fields and language audited; use decision documented before processing.	S1	Not started	Ali Yağız Temizhan
PB-02	Define and document the CSV/JSON input schema and supported field mappings.	Must	§3.4; FR-01–04	Schema documents review_id, account_id, product_id, review_text, rating and timestamp; each field's dependent analyses identified; optional metadata and missing-field behavior specified.	S1	Not started	Ali Yağız Temizhan
PB-03	Implement CSV and JSON dataset upload in batch mode.	Must	FR-01; FR-32	Valid CSV and JSON files can be submitted through the batch upload workflow; imported record count is available; an invalid file format produces a clear error.	S1	Not started	Ali Yağız Temizhan
PB-04	Validate records and report malformed, missing or empty fields.	Must	FR-02; FR-04	Missing, empty and malformed fields are identified; unavailable analyses are skipped or limited with a visible explanation; missing account IDs or timestamps are not inferred.	S1	Not started	Ali Yağız Temizhan
PB-05	Preprocess Turkish review text conservatively.	Must	FR-06	Turkish text is normalized using documented rules; inspected examples confirm that negation and meaning-bearing characters are retained; original text remains available for evidence.	S1	Not started	Ali Yağız Temizhan

B. Text and temporal signals

ID	Backlog item	Priority	Source	Acceptance criteria	Sprint	Status	Assignee
PB-06	Generate sentence embeddings with a Turkish-compatible or multilingual pretrained model.	Must	FR-07	A selected Turkish-compatible or multilingual pretrained model produces embeddings for valid texts; model name/version and processing configuration are recorded.	S1	Not started	Ali Yağız Temizhan
PB-07	Calculate review similarities and configurable thresholds.	Must	FR-08–09	Cosine similarity is computed for supported review pairs; the threshold can be configured; the choice is documented using manually inspected Turkish examples.	S1	Not started	Ali Yağız Temizhan
PB-08	Aggregate review activity in configurable time windows.	Must	FR-11	Given valid product IDs and timestamps, per-product counts are produced for documented configurable windows; missing timestamps disable temporal output with an explanation.	S1	Not started	Gizem Demirer
PB-09	Detect product-level review bursts.	Must	FR-12	A documented rolling-window or local-baseline rule identifies known bursts in a controlled example; detection parameters and flagged windows are inspectable.	S1	Not started	Gizem Demirer
PB-10	Compute near-simultaneous account activity signals.	Must	FR-13	For records with account IDs and timestamps, near-simultaneous multi-account activity is computed; the time window and missing-data limitations are recorded.	S2	Not started	Gizem Demirer

C. Graph and group detection

ID	Backlog item	Priority	Source	Acceptance criteria	Sprint	Status	Assignee
PB-11	Construct the account–product bipartite graph.	Must	FR-15	Available anonymized account IDs and product IDs create a bipartite reviewer–product graph; repeated reviews are handled consistently; missing identifiers prevent unsupported graph claims.	S2	Not started	Bora Özkan
PB-12	Create and constrain weighted account–account edges.	Must	FR-16–17; NFR-07	Account edges are formed from product overlap; available text and timing evidence informs weights; candidate pruning and popular-product handling are documented; absent signals are not fabricated.	S2	Not started	Bora Özkan
PB-13	Detect candidate reviewer communities.	Must	FR-18	A documented community-detection method produces identifiable candidate account groups on a controlled graph; the method and relevant parameters are recorded.	S2	Not started	Bora Özkan

D. Scoring, API and interface

ID	Backlog item	Priority	Source	Acceptance criteria	Sprint	Status	Assignee
PB-14	Compute normalized group-level component signals.	Must	FR-20	For each candidate group, available text, temporal and graph features are normalized and reported separately; unavailable features are explicitly marked.	S2	Not started	Bora Özkan
PB-15	Combine signals into an interpretable 0–100 suspicion score.	Must	FR-21–22; FR-24	Documented configurable weights yield a score between 0 and 100 where sufficient signals exist; groups can be ranked; incomplete-signal scores are distinguished; no fraud-probability claim is made.	S2	Not started	Bora Özkan
PB-16	Generate evidence summaries for each flagged group.	Must	FR-23; NFR-01	Group detail identifies contributing signals and displays representative similar comments, shared products and timing evidence when available; every score can be traced to its signals; unsupported evidence is not shown as fact.	S2	Not started	Bora Özkan
PB-17	Expose upload, analysis initiation and results endpoints.	Must	FR-32	The API supports dataset submission, starting a batch run and retrieving results; a basic end-to-end request sequence is tested with an authorized sample.	S2	Not started	Sinem Şahin
PB-18	Build dashboard list and group-detail views.	Must	FR-33–34	The dashboard lists groups in score order and opens group details; details show score components, supporting evidence and limitations; the human-review disclaimer is visible.	S2	Not started	Sinem Şahin
PB-19	Provide a graph view of a selected group and its products.	Must	FR-35	The selected candidate group and its reviewed products can be inspected in a graph view; identifiers correspond to the analysis output.	S3	Not started	Sinem Şahin

E. Evaluation, quality and safeguards

ID	Backlog item	Priority	Source	Acceptance criteria	Sprint	Status	Assignee
PB-20	Run single-signal and combined evaluation configurations.	Must	FR-26	Text-only, time-only, graph-only and combined configurations are run on comparable records/splits; results are recorded separately without assuming that the combined configuration is better.	S3	Not started	Gizem Demirer
PB-21	Measure supported evaluation metrics and runtime.	Must	FR-27; NFR-07	Runtime is recorded; precision, recall and F1 are reported only at levels with suitable labels; other metrics are used only when meaningful; unavailable metrics have documented reasons.	S3	Not started	Gizem Demirer

ID	Backlog item	Priority	Source	Acceptance criteria	Sprint	Status	Assignee
PB-22	Report real and synthetic results separately.	Must	FR-30	Results on real data and on injected groups are clearly distinguished in every results table and report.	S3	Not started	Gizem Demirer
PB-23	Record reproducible experiment settings and results.	Must	NFR-02; NFR-09; FR-19	Each experiment record includes dataset version/selection, model version, thresholds, windows, weights, seeds when relevant, environment and evaluation procedure; rerun steps are documented.	S1–S3	Not started	Gizem Demirer
PB-24	Apply privacy, access and human-review safeguards.	Must	§1.3; NFR-03–05	Only reviewed/authorized datasets are processed; no live scraping or automated enforcement occurs; public outputs exclude identifying examples; privacy and source restrictions are documented.	S1–S3	Not started	Ali Yağız Temizhan
PB-25	Keep analysis components modular and independently testable.	Must	NFR-06	Data import, preprocessing, text, time, graph, scoring, API and UI logic have clear module boundaries; core components have focused tests; chosen models/algorithms can be replaced with limited changes.	S1–S3	Not started	Sinem Şahin
PB-26	Verify the analyst workflow through the web interface.	Must	NFR-08	An analyst can upload an authorized test dataset, run the analysis and view groups ranked by suspicion score; the interface explains why each group was flagged, states that scores do not prove fraud, and indicates any analysis that could not run because of missing data.	S3	Not started	Sinem Şahin
PB-27	Analyze false-positive examples.	Should	FR-31; §1.3	When false-positive examples are available, representative cases and possible legitimate explanations (e.g., promotions, product launches, customer communities) are documented.	S3	Not started	Gizem Demirer
PB-28	Show import summary and basic time-series charts.	Should	FR-05; FR-36	When the data supports them, the dashboard shows review/account/product counts, field completeness and basic review-activity charts; unavailable summaries are labeled rather than estimated.	S2	Not started	Sinem Şahin
PB-29	Test sensitivity to temporal and similarity thresholds.	Should	FR-14; FR-09	Chosen similarity thresholds and time windows are varied in a recorded experiment; resulting changes in flags or metrics are summarized without claiming an optimal value in advance.	S3	Not started	Gizem Demirer
PB-30	Preserve optional metadata and graph run settings.	Should	FR-03; FR-19	Extra metadata fields do not cause valid imports to fail; graph thresholds and construction parameters are recorded for each relevant analysis run.	S1	Not started	Ali Yağız Temizhan
PB-31	Evaluate controlled synthetic coordinated groups when prerequisites hold.	Conditional	FR-28–29; §1.2	Only when suitable Turkish group labels are insufficient and authorized Turkish background data is available: controlled groups are injected, membership is recorded, recovery is assessed where feasible, and findings are labeled synthetic.	S2–S3	Not started	Gizem Demirer
PB-32	Consider MinHash or equivalent near-duplicate indexing.	Could	FR-10; Table 1	Optional: if measured similarity-search cost warrants it and time permits, an indexing approach is evaluated against the baseline; otherwise it is recorded as deferred.	If time	Not started	Ali Yağız Temizhan
PB-33	Consider learned fusion and SHAP explanations.	Could	FR-25; Table 1	Optional: learned fusion/SHAP is attempted only if reliable labeled training data and time exist and is compared against transparent weighted scoring; otherwise it is left out of the prototype.	If time	Not started	Bora Özkan

F. Course deliverables and project management

ID	Backlog item	Priority	Source	Acceptance criteria	Sprint	Status	Assignee
PB-34	Prepare and submit the Project Proposal (D1).	Must	D1	Proposal submitted by 16 October 2026 and available on the project webpage.	—	Done	Ali Yağız Temizhan
PB-35	Prepare and submit the Project Specifications Report (D2).	Must	D2	Report uploaded to LMS, sent to the supervisor and jury members, submitted via the LMS form and published on the project webpage by 23 October 2026.	S1	Done	Sinem Şahin
PB-36	Prepare and publish Product Backlog Version 1 (D3).	Must	D3	Backlog Version 1 available on the project webpage by 23 October 2026.	S1	Done	Sinem Şahin
PB-37	Prepare and submit the Analysis Report (D4).	Must	D4	Report covering the proposed system, functional and non-functional requirements and system models submitted by 20 November 2026 (LMS upload, email to supervisor and jury, LMS form) and published on the webpage.	S1	Not started	Team
PB-38	Update and publish Product Backlog Version 2 (D5).	Must	D5	Statuses, sprints and assignments updated; Version 2 available on the webpage by 20 November 2026.	S1	Not started	Team
PB-39	Prepare and submit the High-Level Design Report (D6).	Must	D6	Report covering design goals, subsystem decomposition, hardware/software mapping, data management, access control, control flow and boundary conditions uploaded to LMS and published on the webpage by 31 December 2026.	S2	Not started	Team
PB-40	Update and publish Product Backlog Version 3 (D7).	Must	D7	Statuses, sprints and assignments updated; Version 3 available on the webpage by 31 December 2026.	S2	Not started	Team
PB-41	Maintain the project webpage.	Must	NFR-09	All reports and backlog versions are available on https://aliyagizt.github.io/ReviewShield/ by their due dates.	S1–S3	In progress	Ali Yağız Temizhan
PB-42	Prepare the final presentation, poster and demonstration (D8).	Must	D8	A 25-minute recorded presentation in which every member presents a part, including an initial demonstration of the prototype, uploaded to LMS; poster prepared for the demonstration days (4–17 January 2027).	S3	Not started	Team

3. Data-Use Decision Gate

Before using any external dataset, the team will verify its source, usage permissions, required data fields and privacy considerations. Only datasets that meet the project's technical, legal and ethical requirements will be used. Raw data will not be publicly shared without appropriate permission. The final dataset selection will be made after these checks (PB-01).

4. Definition of Done

A backlog item is considered complete when it meets its acceptance criteria, has been tested and is properly documented. Its results should also be demonstrable or reproducible when applicable. ReviewShield's suspicion scores are intended to help users identify groups that may require further investigation, not to prove that reviews are fraudulent.